

Comparative Assembly

How Construction Methods Shape What We Build

CONCEPT

A tabletop installation where the user physically assembles a small wall section, guided by a projector.

The projector overlays real-time LCA data directly onto each physical piece as it is placed: CO₂ cost, labor hours, and material supply chain.

Three construction methods are compared through the same final form — same wall, different process, different environmental footprint.

The data becomes physical. You don't read a chart — you build both versions and feel the difference.

MODULES

Masonry bricks
4 x 2 x 1.5 cm blocks

3DP concrete layers
8 x 2 x 0.5 cm strips

Prefab panels
8 x 4 x 1 cm units

CONNECTION LOGIC

Masonry: gravity stack
±5mm position tolerance

3DP: sequential layers
layer N after N-1 only

Prefab: slot alignment
±10° rotation tolerance

FINAL ASSEMBLED OBJECT

A small wall section (≈25 x 20 cm) built three times, each with a different construction method.

All modules laser-cut or 3D-printed at tabletop scale.
ArUco fiducial markers on underside for camera tracking.

WHITE SPACE

No existing project combines interactive tabletop assembly + projection-mapped LCA data + construction method comparison.

Interaction & Feedback

Human input → System logic → Projection output

INPUT

What human action drives the system?

Physical object placement

Overhead USB webcam reads
ArUco markers on each piece.

No touch. No gesture.
The act of building IS
the input.

Marker gives: piece ID,
position (x,y), rotation (θ)

PROCESSING

FSM + validation rules

5-state FSM controls flow:
IDLE → READY → CHECKING
→ CONFIRMED / ERROR
→ COMPLETE

Validation rule:
Compare detected marker
position/rotation against
target within tolerance.

LCA database lookup per piece.

OUTPUT (Projection)

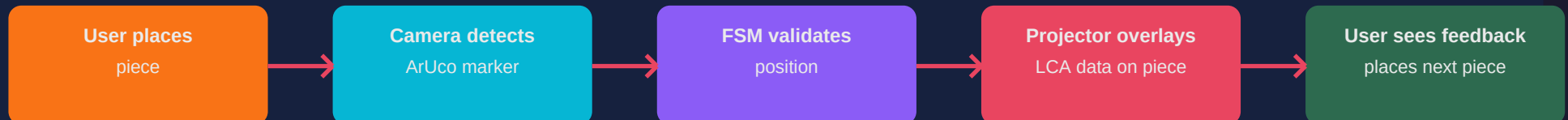
What does projection feedback do?

GUIDE Target outline for next piece
CONFIRM Green pulse on valid placement
INFORM LCA data projected ON the piece:

- CO■ (kg)
- Labor (hours)
- Material origin map

ERROR Red outline + ghost correction
COMPARE Final side-by-side dashboard

THE FEEDBACK LOOP



← loop continues until all pieces placed for current method →

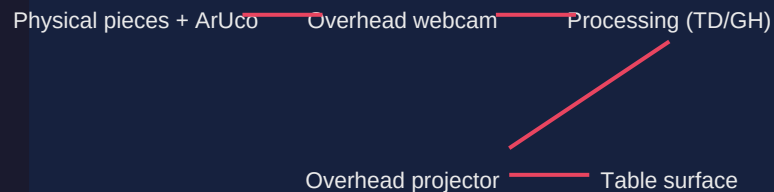
System Architecture

Tech stack, references, and timeline

TECH STACK

Rhino / Grasshopper	Geometry, LCA data, assembly sequence design
TouchDesigner	Runtime: camera input, projection output, FSM
ArUco markers	Fiducial tracking under projector light
HD Projector	Overhead, projection-maps onto table + pieces
USB Webcam	Overhead, feeds marker detection pipeline
OSC Bridge	Grasshopper ↔ TouchDesigner communication

SYSTEM DIAGRAM



← Closed feedback loop: place → detect → validate → project → place →

KEY REFERENCES

Augmented Bricklaying

[Gramazio Kohler Research, ETH Zurich](#)
AR-guided masons placing bricks for complex facades

Steampunk Pavilion

[Fologram + Gwyllim Jahn, Tallinn 2019](#)
Untrained builders assembling curved timber via AR

inFORM

[Hiroshi Ishii / MIT Tangible Media, 2013](#)
Canonical "data you can touch" reference

Material Cultures

[London, 2021–](#)
Bio-based comparison demonstrators

Striatus Bridge

[ZHA + Block Research Group, 2021](#)
3D-printed concrete structural demonstration

EPFL Structural Xploration Lab

[Corentin Fivet](#)
Pavilions from reclaimed components, carbon-driven

TIMELINE

S1 Apr 10: Concept S2 Apr 17: Mapping S3 May 4: FSM S4 May 11: Projection S5 May 18: Integration Finals May 22